Minimum Maximal Domino Packings of the $4 \times n$ Rectangle: An Exact Classification and Enumeration

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Abstract

We study maximal nonoverlapping domino packings of a labeled $4 \times n$ rectangle. An elementary, width-specific proof recovers the known minimum: $m(1) = 1$ and $m(n) = \lceil 4n/3 \rceil$ for $n \geq 2$. A necessary-and-sufficient local grammar describes every minimum packing. A boundary potential restricts its paths to at most three units of nonnegative defect; the recurrent parts are eight horizontal, three-column phase patterns. The number $A(n)$ is eventually constant, linear, or quadratic according to n mod 3, with all exceptions specified. In particular, $A(3k) = 8$ for $k \geq 3$, $A(3k+2) = 50k + 24$ for $k \geq 4$, and $A(3k+1) = 226k^2 + 106k - 392$ for $k \geq 6$. A finite integer-vector identity proves the recurrence for the infinite family. A small certificate, independent graph enumeration, and an independent grammar implementation are supplied. The known minimum, earlier finite counts, and earlier transfer enumeration are explicitly credited; no priority claim is made for the refinements presented here.

1 Problem, conventions, and results

A domino covers two cells sharing an edge. A packing is a set of disjoint domino placements, with no order or labels attached to the dominoes. A *hole* is an uncovered cell. A packing is *maximal* if another domino cannot be added. Equivalently, its holes form an independent set in the cell-adjacency graph. We minimize the number of dominoes among maximal packings and count every packing attaining that minimum.

Rows are labeled $1, 2, 3, 4$ from top to bottom and columns $1, \ldots, n$ from left to right. Reflections and rotations are not identified. Different partitions of the same occupied cells into dominoes are different packings. All boundaries are ordinary rectangle boundaries. In graph language the problem concerns minimum maximal matchings of $P_4 \,\square\, P_n$, where P_j is the path on j vertices.

Theorem 1.1 (Exact minimum)**.** *The minimum number of dominoes in a maximal packing is*

$$m(n) = \begin{cases} 1, & n = 1, \\ \lceil 4n/3 \rceil, & n \geq 2. \end{cases}$$

Theorem 1.2 (Exact count)**.** *With the counting conventions above,*

$$A(3k) = \begin{cases} 7, & k = 1, \\ 10, & k = 2, \\ 8, & k \geq 3, \end{cases}$$

$$A(3k+1) = \begin{cases} 1, 100, 663, 1892, 3652, 5796, & k = 0, 1, 2, 3, 4, 5 \text{ respectively}, \\ 226k^2 + 106k - 392, & k \geq 6, \end{cases}$$

$$A(3k+2) = \begin{cases} 6, 51, 124, 178, & k = 0, 1, 2, 3 \text{ respectively}, \\ 50k + 24, & k \geq 4. \end{cases}$$

Equivalently, the initial values in table 3 through $n = 25$, and

$$A(n) = 3A(n-3) - 3A(n-6) + A(n-9) \qquad (n \geq 26) \tag{1}$$

determine the same sequence.

Theorem 5.2 gives the complete structural classification. Its ingredients are a geometrically defined boundary state, an explicit potential table, and a defect budget of at most three. No accumulated domino count or optimization is needed to apply the classification. Theorem 6.1 makes it especially concrete when $3 \mid n$.

2 Related work and contribution boundary

Minimum maximal matching size is also called a *matching saturation number*. A maximal matching is an independent edge dominating set, and the minimum edge domination number equals the minimum size of a maximal matching. Gyárfás, Lehel, and Tuza's *Clumsy packing of dominoes* [1] is a classical source for the packing problem.

The minimum in theorem 1.1 is already known. Mojdeh and Sadeghi [2, Proposition 2] give width four for $n \equiv 0, 2 \pmod 3$. Klostermeyer and Yeo [3, Theorem 4] give the full result for rectangles with both sides at least two and one side at most ten; their Figure 5 concerns width four. Cibulis and Trump [4, Theorems 1 and 3] prove the area-third lower bound for rectangles with both sides at least two and give equality constructions, again with both sides at least two, when one side is at most thirteen. Their Table 3 already records width-four optimal counts through $n = 20$. Our independent computations reproduce them. The one-column exception is outside those two-dimensional minimum statements.

Transfer enumeration also predates this work. Huntemann and McKay [5, Section 4, Theorem 8] give a bivariate polynomial counting all maximal packings by horizontal and vertical dominoes. Substituting equal weights and extracting the minimum-degree coefficient already determines $A(n)$ algorithmically. Neither the generic transfer idea nor the possibility of exact counting is claimed as new here. The present contribution is the explicit optimal grammar, its bounded-defect and periodic structure, and the proved closed formulas for all lengths with a small reproducible certificate. A search of these sources and related terminology did not locate these precise all-length refinements; that is not a proof of novelty or a claim of priority. Our elementary proof is an independent derivation of the known minimum.

The supplement's `literature.md` records the search scope, primary links, and page-level attribution.

3 An elementary lower bound and constructions

3.1 Directional injections of holes

Write d for the number of dominoes and h for the number of holes, so

$$2d + h = 4n. \tag{2}$$

Let h_L, h_R count holes in the first and last columns, and let d_L, d_R count dominoes meeting those columns.

Assign every hole outside the last column to the domino covering its right neighbor, which is occupied by maximality. This assignment is injective: a horizontal domino has only one exterior cell immediately on its left; two holes immediately left of a vertical domino would be adjacent. No assigned domino meets the first column: a horizontal domino spanning columns $1, 2$ would itself cover the alleged hole. Reflection gives

$$d \geq h - h_R + d_L, \qquad d \geq h - h_L + d_R. \tag{3}$$

An end column has at most two holes. At least two dominoes meet it except in the pattern S: holes in rows $1, 4$, with a vertical domino in rows $2, 3$. Indeed, with at most one hole at least three cells are occupied. With two holes, one domino covers the other two cells only if they are adjacent while the holes are not; this forces S.

Lemma 3.1. *For $n \geq 2$, every maximal packing has $d \geq h$.*

Proof. If either end is not S, (3) gives $d \geq h$. Otherwise, suppose $d < h$. Both ends are S, and the inequalities force $d = h - 1$. Both injections are bijections onto the dominoes not meeting the corresponding end. Consequently:

(L) every domino not meeting the first column has a hole immediately to its left;

(R) every domino not meeting the last column has a hole immediately to its right.

A hole beside a vertical domino can be beside either cell; for a horizontal domino it must be beside the exterior end.

The left central domino forces a hole in row 2 or 3 of column 2 by (R). Rows $1, 4$ of column 2 are occupied, since their left neighbors are holes. They cannot both belong to vertical dominoes, which would occupy all of column 2. They cannot both start horizontal dominoes: the middle cells would then have to be holes, since any domino covering them would violate (L), but those holes would be adjacent. Thus exactly one outer cell starts a horizontal domino. Reflecting rows if needed, column 2 has a horizontal domino starting in row 1, a hole in row 2, and a vertical domino in rows $3, 4$. For $n = 2$ this already puts a domino outside the board.

In column 3, row 1 belongs to the incoming horizontal domino. Row 4 must be a hole: no horizontal domino can enter from the covered cell to its left, and a new horizontal domino or a vertical domino in rows $3, 4$ would violate (L). Row 3 must therefore be occupied. It cannot belong to a horizontal domino, by (L) and the preceding vertical domino; thus rows $2, 3$ form a vertical domino. For $n = 3$ this contradicts the required right-end pattern S.

For $n \geq 4$, condition (R) for the horizontal domino in row 1 forces a hole at $(1, 4)$. Condition (R) for the column-3 vertical domino forces a hole at $(2, 4)$ or $(3, 4)$; only $(3, 4)$ is possible, since holes cannot be adjacent. Cell $(2, 4)$ must be occupied, but both vertical neighbors are holes, its left neighbor is in a vertical domino, and a horizontal domino starting there would violate (L). This contradiction proves the lemma. □

By (2), the lemma gives $3d \geq 4n$. For $n = 1$ a single central vertical domino is maximal, and zero dominoes is not. This proves all lower bounds in theorem 1.1.

3.2 Explicit optimal words

Let F be the fully covered one-column board with vertical dominoes in rows $1, 2$ and $3, 4$. Define width-three blocks by local coordinate pairs:

$$P = \{(2,1)(3,1),\ (1,2)(1,3),\ (3,2)(4,2),\ (2,3)(3,3)\},$$
$$B = \{(2,1)(2,2),\ (4,1)(4,2),\ (1,2)(1,3),\ (3,2)(3,3)\},$$
$$Q = \{(4,1)(4,2),\ (2,1)(3,1),\ (1,2)(2,2),\ (2,3)(3,3)\}.$$

Each pair of coordinates denotes one domino. Each block has four dominoes.

For $n = 1, 2, 3, 4, 5$, use $S, SF, SFS, SFSF, SFSFS$, respectively. For $k \geq 0$, use

width	word	dominoes
$6 + 3k$	PB^kQ	$8 + 4k$
$7 + 3k$	PB^kQF	$10 + 4k$
$8 + 3k$	PB^kQFS	$11 + 4k$.

(4)

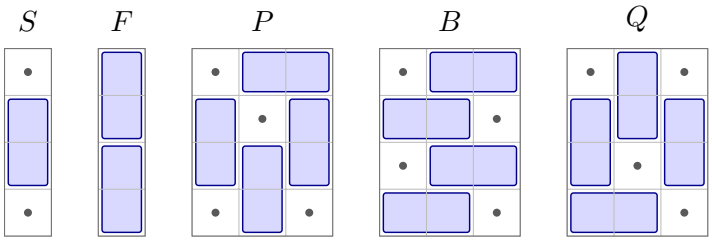


Figure 1: Construction blocks. Rounded blue rectangles are dominoes and black dots are holes. Concatenation places blocks side by side without dominoes crossing between them.

First-column and last-column hole sets are, respectively,

$$P : \{1,4\},\{4\}; \qquad B : \{1,3\},\{2,4\}; \qquad Q : \{1\},\{1,4\}.$$

They are disjoint at every join above, including $P|Q$ when $k = 0$. The fully occupied F makes its joins automatic. Within every block, holes are also nonadjacent. The words are therefore maximal and attain the bounds. This proves theorem 1.1 for every length without a computational proof step.

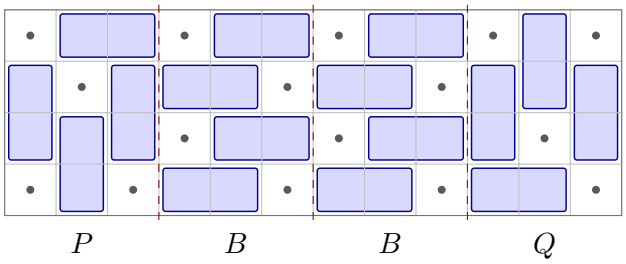


Figure 2: The optimal 4×12 packing $PBBQ$, with sixteen dominoes. Dashed lines mark block joins.

4 A precise column model

Represent a subset R of rows by the binary mask $\sum_{r\in R} 2^{r-1}$. Masks $5, 9, 10$ represent $\{1,3\},\{1,4\},\{2,4\}$. In formulas, & and | denote bitwise intersection and union, and $|x|$ counts set bits. The independent hole masks are

$$\mathcal{U} = \{0,1,2,4,5,8,9,10\}.$$

After processing a column, the state (h, u) records horizontal dominoes starting there and continuing into the next column in h, and holes in u. Thus, before the next column is processed, h is its set of already occupied incoming cells. The state set is

$$\mathcal{S} = \{(h,u) : 0 \le h \le 15,\ u \in \mathcal{U},\ h \,\&\, u = 0\}. \tag{5}$$

It has sixty states. The initial state is $s_0 = (0, 0)$: no incoming dominoes and no holes in a fictitious preceding column. Terminal states are $(0, u)$, so no domino exits the rectangle.

A transition $s = (h, u) \to t = (k, v)$ processes one column: h are incoming cells, k outgoing cells, v holes, and the remaining cells are covered vertically. Its exact conditions are

$$\begin{gathered} h \,\&\, (k \mid v) = 0, \qquad u \,\&\, v = 0, \\ R = 15 \operatorname{xor} (h \mid k \mid v) \in \{0,3,6,12,15\}. \end{gathered} \tag{6}$$

The disjointness $k \,\&\, v = 0$ is already part of $t \in \mathcal{S}$. The five listed residual masks are exactly the subsets of a four-vertex path with a perfect matching, which is unique. Hence each ordered pair of states supports at most one transition. Its new-domino count is

$$c(s,t) = |k| + \frac{|R|}{2} = \frac{4 - |h| - |v| + |k|}{2}. \tag{7}$$

Horizontal dominoes are counted when they start, not when they arrive. There are 304 transitions.

Proposition 4.1 (Exact encoding)**.** *Length-n paths from s_0 to a terminal state are in bijection with maximal packings. Their total transition cost is their domino count.*

Proof. A packing uniquely determines outgoing and hole masks in every column. Its incoming cells, outgoing cells, holes, and vertical pairs satisfy (6). Holes within a column are independent because $v \in \mathcal{U}$; holes across successive columns are independent exactly when $u \,\&\, v = 0$. These are all possible hole adjacencies. Conversely, a path gives each outgoing domino its incoming cell in the next column and gives unique vertical dominoes from the residual mask. Initial and terminal conditions exclude overhang at both ends. Every cell has one status, and the hole conditions give maximality. These constructions are inverse. Formula (7) counts every domino exactly once. □

Thus different partitions of occupied cells are distinguished, while placement orders are not. No multiplicity factor or symmetry quotient is needed.

5 Optimality as a bounded local defect

Give each state the potential $p(h,u)$ in table 1. A dash means that the pair is not a state. This potential is a boundary correction making local excess over the four-dominoes-per-three-columns density nonnegative.

Table 1: Boundary potential. Row and column labels are decimal binary masks, not row numbers.

$h\backslash u$	0	1	2	4	5	8	9	10
0	0	0	0	0	0	0	-1	0
1	3	–	1	3	–	2	–	1
2	3	2	–	3	2	2	2	–
3	4	–	–	4	–	4	–	–
4	3	2	3	–	–	2	2	2
5	4	–	3	–	–	4	–	2
6	4	3	–	–	–	3	2	–
7	4	–	–	–	–	4	–	–
8	3	2	3	1	1	–	–	–
9	4	–	4	4	–	–	–	–
10	4	4	–	3	2	–	–	–
11	4	–	–	4	–	–	–	–
12	4	4	4	–	–	–	–	–
13	4	–	4	–	–	–	–	–
14	4	4	–	–	–	–	–	–
15	4	–	–	–	–	–	–	–

For a legal transition define its *defect*

$$\delta(s,t) = 3c(s,t) - 4 + p(s) - p(t). \tag{8}$$

Lemma 5.1 (Finite local certificate)**.** *The displayed potential has the following properties.*

1. *Every legal transition has nonnegative defect. The counts at defects* $0, 1, 2, 3, 4$ *are* $111, 99, 59, 20, 15$*, respectively.*

2. *In the zero-defect graph the only states both reachable from* $(0, 0)$ *and able to reach* $(0, 9)$ *are* $(0, 0)$ *and* $(0, 9)$*. The only edge between them is* $(0, 0) \to (0, 9)$*.*

3. *The only nontrivial strongly connected components are the eight directed three-cycles in table 2. There are no self-loops.*

Finite verification. Enumerate (5), test (6), and evaluate (7) and (8) using the table. This tests exactly 304 integer inequalities and gives the histogram. For the other assertions, compute reachability by successive insertion of out-neighbors. The component containing s is the intersection of its forward and backward reachable sets. Every input is displayed in the paper. The executable `verify.py` implements these operations and compares their output with `certificate.json`; an independent implementation generates transitions by physically filling a column. Both checks passed. This is an exhaustive finite proof step, not an inference from a sample of board lengths. □

For a complete path with d dominoes, total defect b, and final state t, telescoping gives

$$b = 3d - 4n - p(t), \qquad 3d - 4n = b + p(t), \tag{9}$$

because $p(s_0) = 0$. Terminal potentials are zero except $p(0, 9) = -1$. This also gives an alternative lower bound: negative $3d - 4n$ would require $b = 0, t = (0, 9)$, possible only for the single-column packing S by the finite certificate.

Theorem 5.2 (Complete optimal-packing grammar)**.** *For* $n = 1$ *the unique optimum is* S*. For* $n \geq 2$*, every optimum is generated exactly once by the following rules, and every generated packing is optimal.*

1. *Start at* $(0, 0)$ *with budget used* $b = 0$*.*

2. *At each column take a transition satisfying* (6)*, place its indicated dominoes and holes, and add* (8) *to* b*. Never allow* $b > 3$*.*

3. *After* n *columns, finish at* $(0, u)$ *and require*

$$0 \leq b + p(0, u) \leq 2. \tag{10}$$

Equivalently, write $r_n = (-n \bmod 3) \in \{0, 1, 2\}$*. The final budget is* r_n *if* $u \neq 9$*, and* $r_n + 1$ *if* $u = 9$*.*

Proof. An accepted path satisfies $0 \leq 3d - 4n \leq 2$ by (9), so its integer domino count must be $\lceil 4n/3 \rceil$, the optimum. Conversely, an optimum at $n \geq 2$ has $3d - 4n = r_n \in \{0, 1, 2\}$, so its total defect is exactly the stated terminal budget, at most three. Every defect is nonnegative; no prefix exceeds the total, so no optimum is discarded. Uniqueness follows from theorem 4.1. At $n = 1$ direct inspection gives S as the sole optimum. □

The grammar uses local tests and four budget values, not minimization over packings. Every optimal path has at most three positive-defect transitions. Arbitrarily long zero-defect stretches have the structure described next.

Table 2: All recurrent zero-defect patterns. Each row is a directed cycle, with an implicit arrow from its third state back to its first.

First state	Second state	Third state
$(0, 5)$	$(5, 10)$	$(10, 0)$
$(0, 10)$	$(10, 5)$	$(5, 0)$
$(1, 4)$	$(4, 10)$	$(10, 1)$
$(1, 10)$	$(10, 4)$	$(4, 1)$
$(2, 4)$	$(4, 9)$	$(9, 2)$
$(2, 5)$	$(5, 8)$	$(8, 2)$
$(2, 8)$	$(8, 5)$	$(5, 2)$
$(2, 9)$	$(9, 4)$	$(4, 2)$

6 Periodic bulk and the eight-packing cases

All edges of these cycles place only horizontal dominoes. Each row repeats a hole and a horizontal domino every three columns. Let $a_i \in \mathbb{Z}/3\mathbb{Z}$ be row i's hole residue. Maximality between rows says exactly $a_i \neq a_{i+1}$. Thus the recurrent phases are proper three-colorings of a four-vertex path, up to simultaneous phase translation. Their number is $3 \cdot 2^3/3 = 8$. The table identifies them with the eight components, and figure 3 displays them. The finite check excludes other recurrent patterns; it does not assert that every entire optimal board is periodic.

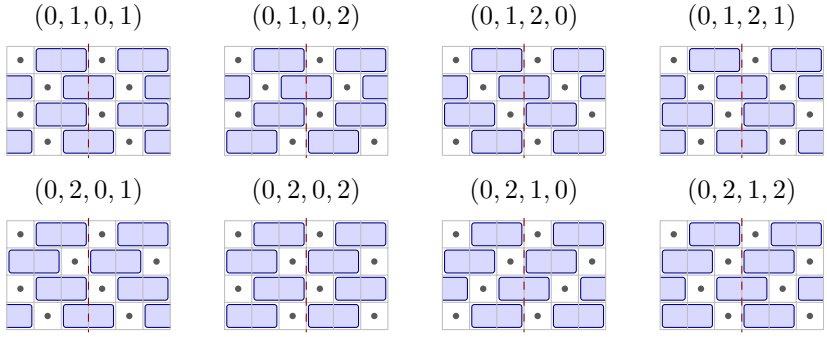


Figure 3: The eight phases normalized by $a_1 = 0$. Two periods are shown; labels give hole residues, with the left displayed column at residue 0. Dominoes cut by display edges continue into neighboring periods. These are interior motifs, not wrapped boards; the grammar still imposes ordinary rectangular boundaries.

Form a graph with vertices (s, b) for $s \in \mathcal{S}$, $0 \leq b \leq 3$, and edges

$$(s, b) \longrightarrow (t, b + \delta(s, t)) \quad \text{if } b + \delta(s, t) \leq 3. \tag{11}$$

Retain exactly the vertices reachable from $(s_0, 0)$ and able to reach a terminal satisfying (10). Denote this trimmed graph by $\mathcal{G}$. This prescription fully specifies it without an unexplained state list. The finite checks give 114 vertices and 327 edges. Its sixteen cyclic components are all directed three-cycles, and every accepted path meets at most three. They are budget-layer copies of the eight phase types. These facts are checked using forward/backward reachability and longest paths in the acyclic graph of components, in both implementations.

Every optimum is consequently a finite sequence of transient pieces and at most three repeating sections from those phases. Repeating sections have arbitrary nonnegative numbers of extra three-column turns; the grammar specifies all entry, exit, boundary, and defect conditions. There are only finitely many routes through the acyclic component graph. Each three-cycle contributes a factor $1/(1 - z^3)$, so the counting generating function has denominator dividing $(1 - z^3)^3$. This explains the eventual degree-two bound as well as the period three.

Corollary 6.1. *For $n = 3k$ with $k \geq 3$, the complete list is*

$$B^k, \qquad PB^{k-1}, \qquad B^{k-1}Q, \qquad PB^{k-2}Q,$$

and their four top-to-bottom row reflections.

Proof. The listed hole sets show maximality at every join, and each word has $4k$ dominoes. All eight are distinct. A central vertical domino in the first column records the presence of P, and one in the last column records the presence of Q. Within any of these four choices, reflection changes the first-column hole mask 5 to 10 when P is absent; when P is present, it moves that block's row-1 horizontal domino to row 4. The count $A(3k) = 8$ proved below therefore gives completeness. □

7 Exact enumeration and the infinite-family certificate

7.1 A finite system counting the grammar

For each state s and budget $b \in \{0, 1, 2, 3\}$ define

$$f_0(s, b) = \begin{cases} 1, & (s, b) = (s_0, 0), \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

$$f_{j+1}(t, b) = \sum_{s \to t} f_j(s, b - \delta(s, t)), \tag{13}$$

with values at budgets outside $\{0, 1, 2, 3\}$ set to zero. Put

$$C_j = \sum_{u \in \mathcal{U}} \sum_{\substack{0 \leq b \leq 3 \\ 0 \leq b + p(0,u) \leq 2}} f_j((0, u), b). \tag{14}$$

This fully specified recurrence has a fixed number of states per column and never enumerates individual packings. The encoding and grammar theorems prove its multiplicities and $C_j = A(j)$ for $j \geq 2$.

There is an important boundary distinction. $C_0 = 1$ counts the empty path. At length one this grammar counts F, not the true minimum S. Both sets have size one, so $C_1 = A(1) = 1$ numerically. The structural rule at length one must still be replaced as stated in theorem 5.2; no packing-preserving correspondence is claimed there.

7.2 The all-length proof

Order the vertices of $\mathcal{G}$ by increasing (b, h, u) and let T be its 114×114 adjacency matrix, with sources indexing rows. Let v_0 be the row vector with one at $(s_0, 0)$ and zero elsewhere, $v_j = v_0 T^j$, and a the column indicator of accepted terminal vertices. Then $C_j = v_j a$. All these objects are determined by the formulas and table in the paper, without any optimization oracle.

The exact finite certificate is

$$\boxed{v_{26} - 3v_{23} + 3v_{20} - v_{17} = 0 \quad \text{in } \mathbb{Z}^{114}.} \tag{15}$$

The supplementary certificate supplies the four vectors. The verifier rebuilds the graph, recomputes the vectors, compares every supplied entry, and checks all 114 equalities with exact integers. The independent grammar implementation obtains the same identity from a different column-generation procedure.

For $n \geq 26$, multiply (15) by T^{n-26} and then by a. This gives

$$C_n - 3C_{n-3} + 3C_{n-6} - C_{n-9} = 0.$$

All involved lengths are positive, so this is (1). This matrix argument is the bridge from the finite check to the infinite family. Agreement of finitely many scalar counts would not provide it.

The initial values follow from (12)–(14). The formulas in theorem 1.2 agree through $n = 25$ and satisfy the same recurrence thereafter: a third finite difference of a polynomial of degree at most two is zero. At the first recurrence in each residue, the earliest terms are 18 (residue 0), 19 (residue 1), and 17 (residue 2), within their stated polynomial regimes. Induction proves theorem 1.2.

Table 3: Checked minima and counts. Independent graph recursion checked these values and continued through 40. Raw, unmemoized enumeration also checked $1 \le n \le 9$. Counts through 20 agree with the previously published Cibulis–Trump table.

n	$m(n)$	$A(n)$	n	$m(n)$	$A(n)$	n	$m(n)$	$A(n)$
1	1	1	10	14	1892	19	26	8380
2	3	6	11	15	178	20	27	324
3	4	7	12	16	8	21	28	8
4	6	100	13	18	3652	22	30	11424
5	7	51	14	19	224	23	31	374
6	8	10	15	20	8	24	32	8
7	10	663	16	22	5796	25	34	14920
8	11	124	17	23	274	26	35	424
9	12	8	18	24	8	27	36	8

7.3 Rational generating function

Let $A(z) = \sum_{n \ge 1} A(n) z^n$. Then

$$A(z) = \frac{N(z)}{(1 - z^3)^3}, \tag{16}$$

where

$$\begin{aligned} N(z) = {} & z + 6z^2 + 7z^3 + 97z^4 + 33z^5 - 11z^6 \\ & + 366z^7 - 11z^8 - z^9 + 202z^{10} - 47z^{11} + 7z^{12} \\ & - 135z^{13} + 11z^{14} - 2z^{15} - 147z^{16} + 12z^{17} \\ & + 56z^{19} - 4z^{20} + 20z^{22} - 8z^{25}. \end{aligned}$$

Multiplying the series by $1 - 3z^3 + 3z^6 - z^9$ gives this numerator from the initial values; the recurrence makes all coefficients at degree 26 and above zero. Thus this rational function follows from the certificate, not a guessed fit.

8 Verification, reproducibility, and limitations

All computations use exact integers. The supplement requires Python 3.9 or later and only its standard library. No remote service, solver, or private data is needed. Run these commands from the extracted directory:

```
python verify.py --independent
python independent_grammar_audit.py
python build_figures.py
pdflatex -interaction=nonstopmode -halt-on-error -jobname=minimum_maximal_domino_packings_4xn
     main.tex
pdflatex -interaction=nonstopmode -halt-on-error -jobname=minimum_maximal_domino_packings_4xn
     main.tex
```

Each compilation command is one shell command; its typeset line may wrap. A TeX distribution with the packages in `main.tex`, including TikZ, is required only to rebuild the PDF. Figure regeneration is optional because the editable figure sources are included.

Certificate. `verify.py` contains the potential table and regenerates the legal transitions, the trimmed graph, the eight base cycles, the sixteen layered cyclic components, and the maximum of three cyclic components on an accepted route. It recomputes the four certificate vectors, initial counts, and numerator, and compares the resulting finite objects with `certificate.json`. It also tests the explicit constructions for $n = 1, \ldots, 200$; their all-length validity is proved independently above.

Independent matching enumeration. `independent_check.py` constructs the grid as an explicit graph and takes its first remaining vertex. It either leaves that vertex unmatched or pairs it with one remaining neighbor. A forbidden mask records remaining neighbors of earlier unmatched vertices; a forbidden vertex may not be left unmatched. Terminal branches are exactly maximal matchings, since their unmatched sets are independent. Every matching chooses a unique branch at each step, eliminating order duplicates. Memoization shares identical remaining subproblems without changing the recurrence. This method uses neither column states nor the potential. It agrees on minimum and count through $n = 40$. Its unmemoized version also agrees through $n = 9$, using 12,291,629 recursion calls at $n = 9$.

Independent grammar audit. `independent_grammar_audit.py` recursively fills a column's first free cell as a hole, an outgoing horizontal domino, or a vertical domino. It reproduces all states and transitions, potential inequalities, phase patterns, trimmed components, and the full vector identity. It imports neither the transfer model nor the certificate verifier. This is an implementation cross-check; the infinite result follows from the encoding and the matrix proof, not from finite sample agreement.

Verification scope. The minimum has an elementary proof. The structural and counting theorems use the displayed finite checks together with proved encoding and matrix arguments. The programs were executed, not merely written. This is a computer-assisted proof with a reproducible certificate; no Lean formalization was compiled and no independent human peer review is claimed. Independent implementations here mean different algorithms within the same investigation. Published finite counts are corroboration, not proof for unbounded n.

The source was compiled to the accompanying PDF and rendered pages were inspected for legibility and layout. The supplement includes LaTeX, editable TikZ figures, programs, certificate, result data, the paper, and a README with commands. It excludes third-party research PDFs and machine-specific paths; the references use public source links.

AI disclosure. This manuscript, its arguments, computations, code, diagrams, and literature audit were produced by GPT-6 Astra, an AI system. Separate internal passes handled literature, elementary proof and constructions, and independent computation and auditing. Such internal cross-checking is not external peer review. The claims are supported by the explicit arguments and reproducible artifacts, not by the identity of the tool producing them.

A Compact description of the finite proof check

The following pseudocode states the essential check without searching for an optimum. It uses only the displayed potential and transition rules. The operation `closure` repeatedly adds neighbors until its finite set stops changing.

```
S = [(h,u) for h in 0..15 for u in {0,1,2,4,5,8,9,10}
             if (h & u) == 0]
E = []
for (h,u) in S:
  for (k,v) in S:
    if h & (k|v) or u & v: continue
    R = 15 xor (h|k|v)
    if R not in {0,3,6,12,15}: continue
    c = popcount(k) + popcount(R)/2
    d = 3*c - 4 + p(h,u) - p(k,v)
    assert d >= 0
    E.append(((h,u),(k,v),d))
L = [(s,b) -> (t,b+d) for (s,t,d) in E for b in 0..3
                              if b+d <= 3]
F = {( (0,u),b ): 0 <= b+p(0,u) <= 2, 0 <= b <= 3}
V = closure({((0,0),0)},L) intersect closure(F,reverse(L))
G = all edges of L with both endpoints in V
assert len(V) == 114 and len(G) == 327
v[0] = unit vector at ((0,0),0), ordered by (b,h,u)
for j in 0..25:
  v[j+1] = zero vector
  for edge i -> k in G: v[j+1][k] += v[j][i]
assert v[26] - 3*v[23] + 3*v[20] - v[17] == zero vector
```

The executable verifier additionally checks all certificate entries and the component claims. Trimming changes no accepted path: every vertex on one is reachable from the start and can reach an accepted terminal. Conversely, trimming cannot create paths. Thus the identity holds on exactly the graph relevant to enumeration.

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